

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 9: Air Masses & Frontal Systems
(Warm, Cold, Occluded)

Syllabus Source: CAE Oxford (pp. 279-322) | Pass Target: >= 90% AviationExam

1. AIR MASSES & SOURCE REGIONS

An air mass is a large body of air whose temperature and humidity characteristics are relatively uniform over thousands of kilometers:

- Source Classifications: Arctic (A), Polar (P), Tropical (T); subdivided into Maritime (m) (moist) and Continental (c) (dry). E.g. mP = Maritime Polar (cool, moist, unstable); cT = Continental Tropical (hot, dry, unstable).

2. FRONTAL SYSTEMS COMPARISON

Frontal Parameter	Warm Front (Warm air replaces cold)	Cold Front (Cold air replaces warm)
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Frontal Slope & Speed	Shallow slope (1:150). Slower speed (~15 to 20 kt).	Steep slope (1:50). Fast speed (~25 to 40 kt).
Cloud Sequence on Approach	Ci -> Cs (halo) -> As (watery sun) -> Ns (continuous rain).	Cu building rapidly into Cb and squall lines.
Precipitation Type	Continuous, widespread rain or snow. Extensive low Stratus.	Heavy showers, hail, thunderstorms, brief intense downpour.
Wind Behavior during Passage	Veers (clockwise in N Hemisphere) and decreases slightly.	Sharp, violent VEER (e.g. 210° to 300°), gusty with squalls.
Post-Frontal Weather	Warm sector: High dew point, poor visibility, mist, low drizzle.	Cold air: Rapid clearing, marked temperature DROP, excellent visibility!

3. OCCLUDED FRONTS (COLD VS WARM OCCLUSION)

Because cold fronts move faster than warm fronts, the cold front eventually catches up with the warm front, lifting the warm sector completely off the ground:

- Cold Occlusion: The air behind the advancing cold front is COLDER than the retreating cold air ahead of the warm front. The advancing cold wedge burrows beneath all air masses.
- Warm Occlusion: The air ahead of the warm front is colder than the air behind the cold front. The advancing cold air rides up over the very cold retreating air.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Warm Front vs Cold Front Flight

Warm Front (Shallow slope 1:150, slow 15-20 kt): - Cloud Sequence: Cirrus (Ci, 600 NM ahead) -> Cirrostratus (Cs, halo) -> Altostratus (As) -> Nimbostratus (Ns, continuous rain and low ceilings). - At Passage: Pressure stops falling, temperature RISES, wind VEERS, poor visibility improves in warm sector. Cold Front (Steep slope 1:50, fast 25-40 kt): - Cloud Sequence: Cumulus (Cu) rapidly developing into Cumulonimbus (CB) squall line, heavy rain showers, severe turbulence. - At Passage: Sharp wind VEER, barometric pressure jumps upward, temperature DROPS, excellent visibility.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the typical cloud sequence observed ahead of an approaching WARM FRONT?**

- [A]: Cb, TCU, Cu, St
- [B]: CIRRUS (Ci), CIRROSTRATUS (Cs), ALTOSTRATUS (As), and NIMBOSTRATUS (Ns)
- [C]: Stratus, fog, clear sky
- [D]: Altocumulus lenticularis

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The gentle slope (1:150) of a warm front pushes warm air aloft up to 600 NM ahead of the surface front, creating a descending cloud shield: Ci -> Cs -> As -> Ns.

Q2: What happens to the wind direction in the Northern Hemisphere during the passage of an active COLD FRONT?

- [A]: Backs (turns counter-clockwise)
- [B]: SHARPLY VEERS (turns clockwise, e.g. from Southwest to Northwest)
- [C]: Remains perfectly constant
- [D]: Drops to calm

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Cold front passage is marked by a sharp clockwise wind shift (veering) as the trough line crosses, typically shifting from SW to WNW or NW.

Q3: What characterizes the weather in the COLD AIR MASS immediately behind an active cold front?

- [A]: Continuous low stratus and poor visibility
- [B]: RAPID CLEARING with isolated convective showers, dropping temperature, and EXCELLENT VISIBILITY
- [C]: Warm sector mist
- [D]: Severe fog

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The cold, dense polar air is dry and clean, resulting in rapid pressure rise, gusty winds, dropping dew points, and superb visibility outside convective showers.